

History-Driven Suppression Reshapes Visuospatial Tuning Across the Visual Hierarchy

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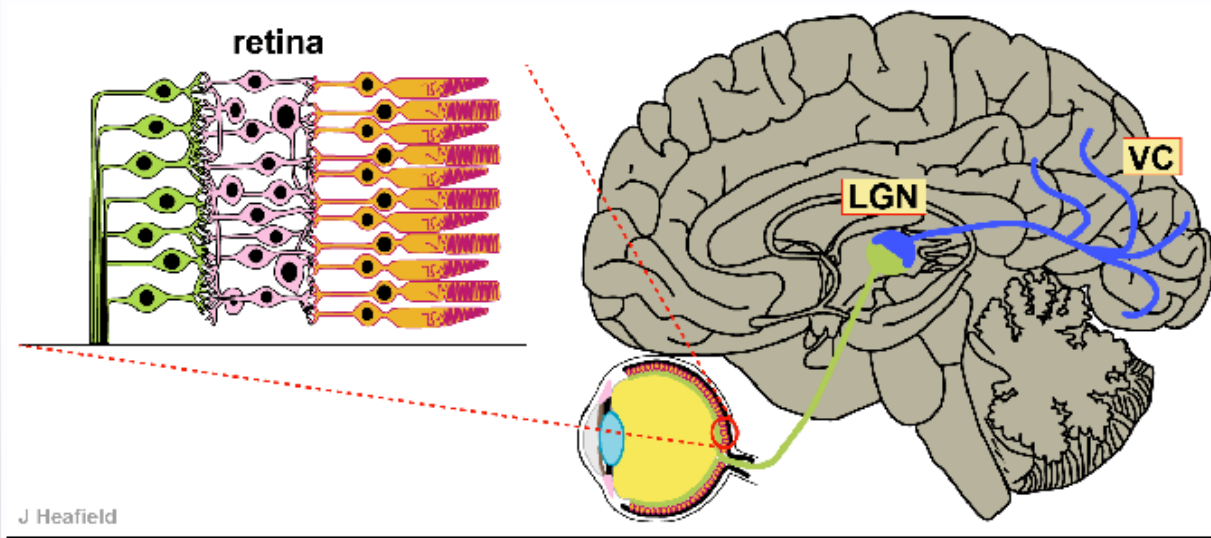
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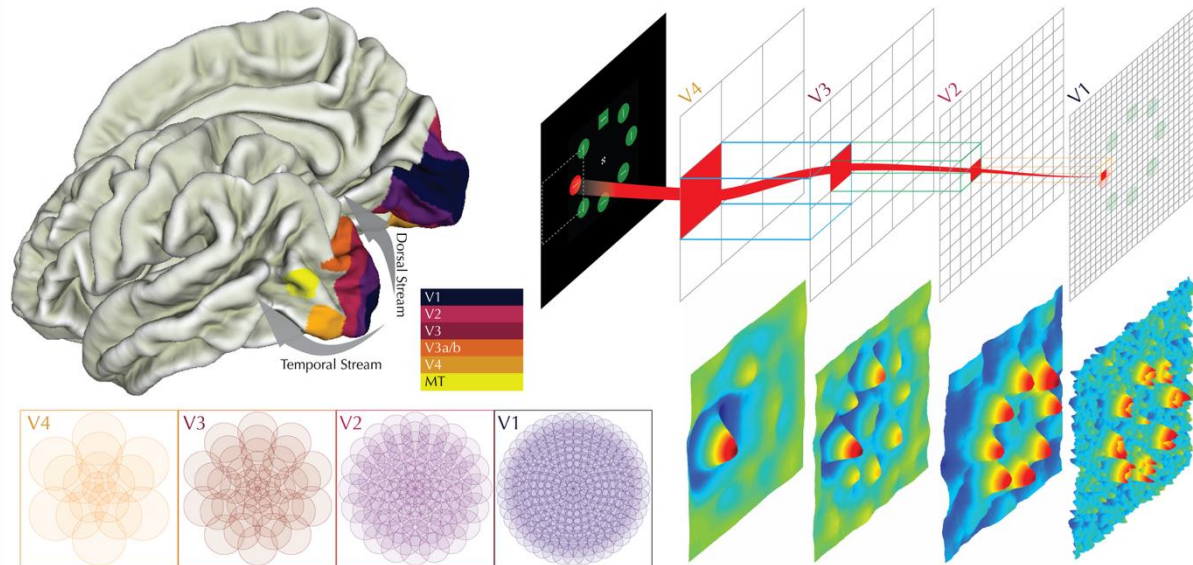
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What is attention again? (in vision)



- The flow of information from our senses through our brain gives rise to a phenomenon known as attention
- Visual scenes are not processed in whole, but rather subdivided such that some of our perceptual input receives greater representation in the brain
- Information is lost somewhere on the way
- We all here love and study attention



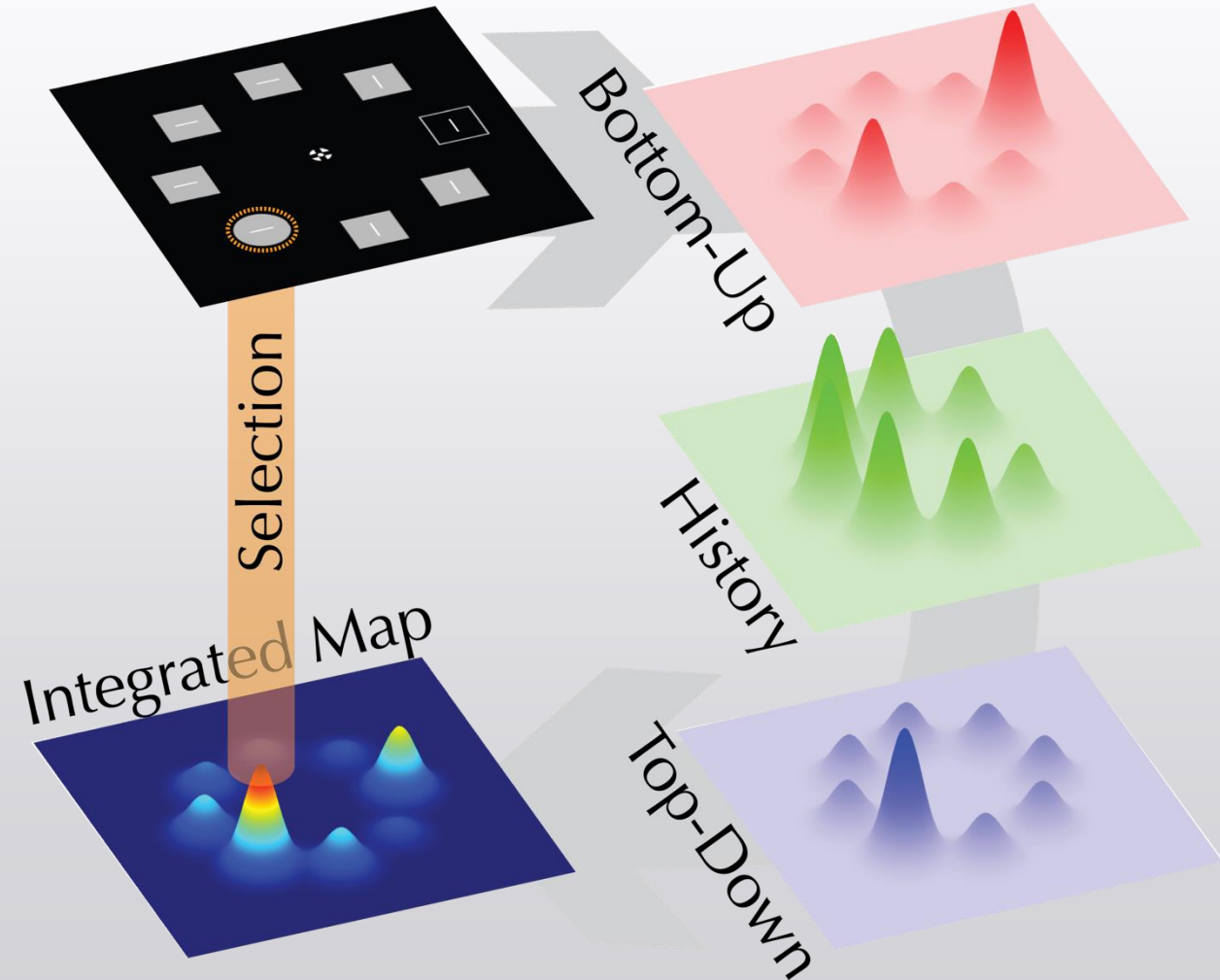
Salience

- Salience as a major driver of attention
- From the earliest stages of visual processing, salience is influential
 - V1 salience map
- Great when search target is salient
- Bad when it isn't



Statistically Learned Suppression

- Various scenarios where the detrimental effects of salience can be mitigated
- **Importantly, cuing an upcoming distractor location or feature does nothing to mitigate interference**
- Alternatively, when irrelevant salient distractors are regularly presented in the same spatial location then capture is reduced, indicating experience is an important tool for handling distraction
- Open question how the brain handles distraction at these locations better
- One theory is that the bottom up input at these areas is inhibited, such that the gain needed to win processing at higher cortical levels increases



- Studying the propagation of spatial salient information lends itself well to population receptive field mapping

Neuron
Article

Attraction of Position Preference by Spatial Attention throughout Human Visual Cortex

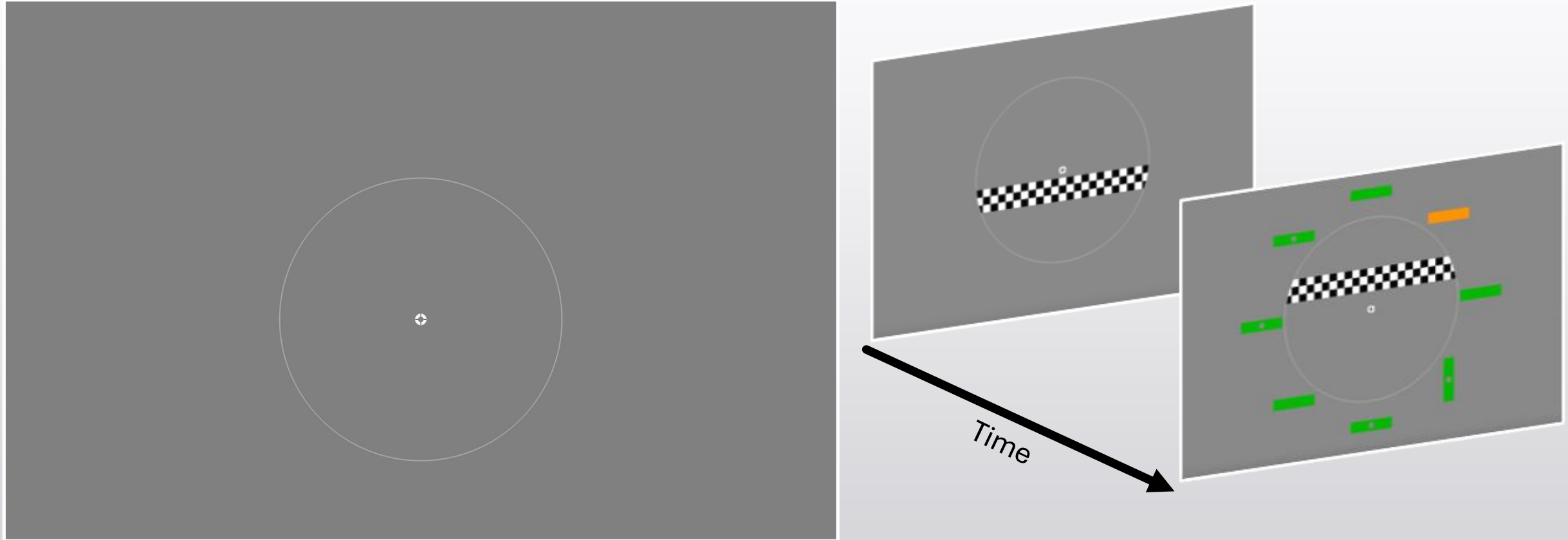
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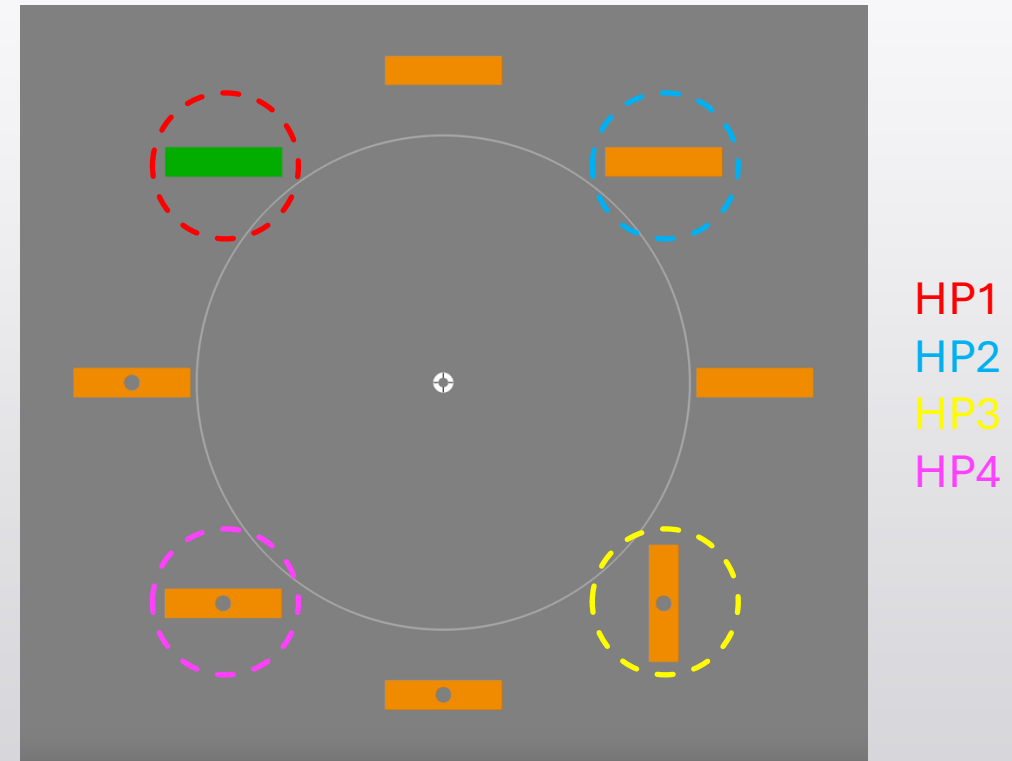
The Paradigm



- We used PRF mapping while participants concurrently performed a visual search task in the scanner
- 7 tesla Philips scanner
- 28 participants, 2 sessions each

Methods stuff

- Blocked HP distractor locations
- Observed how PRF fits changed as a function of the current high probability distractor location
- (give history on shifting PRFs as a tool for measuring attention)
- Could be that because distractors were presented more frequently at one location, then fields shift towards that location because there is more bottom-up attentional representation there
- Alternatively, could shift away, indexing a reduction in receptivity in these frequent distractor locations
- Importantly, targets were presented at each location the same number of times, so unrelated to target processing



- Prf map on 3d brain and flat brain to show fitting worked

- Shifting receptive fields across conditions. Could do arrow plots

- Shift by processing area, show shift is greatest in later areas

Interpreting the shift

- What does a receptive field shift mean?

○ The end